2 Verification Cases

For the verification of the numerical solvers, a few test cases of oscillatory motion are considered. Firstly, a case of fluid flow over an oscillatory flat plate is considered. Thereafter, the case of oscillatory round tube has also been taken into account and the results obtained from the present numerical simulations have been compared with the respective analytical results.

2.1 Oscillatory Flat Plate

The oscillatory flat plate over a still fluid is known as Stoke's second problem where one generally not interested in the complete solution but only the periodic steady state solution. For this problem, initially a semi-infinite body of fluid and is bounded by

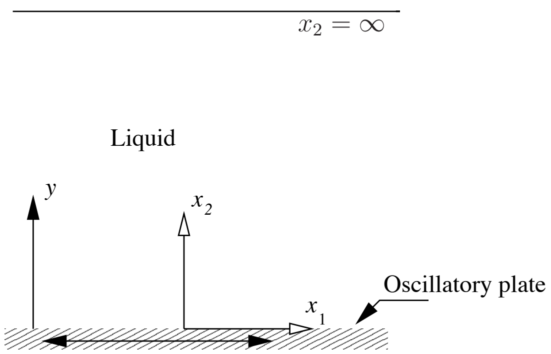

Figure 1: A schematic view of the flow over an oscillatory plate.

a flat plate as shown in Fig. 1 and both fluid and the flat plate are a rest at $t = 0$. Consequently, the oscillatory motion of the flat plate in x direction is initiated such that the plate is displaced by an amplitude A_0 and an angular frequency of $\omega = 2\pi f$. That is the displacement x and velocity U of the plate are given by:c

$$\begin{aligned} x(t) &= x_0 \sin(\omega t) \\ U(x_2 = 0, t) &= A_0 \omega \cos(\omega t) = U_0 \cos(\omega t) \end{aligned} \tag{1}$$

where f denotes the frequency of oscillations.

The analytical solution of this is obtained [1, 2] by integrating the following equation for unsteady-state flow:

$$\frac{\partial U}{\partial t} = \nu \frac{\partial^2 U}{\partial x_2^2} \tag{2}$$

On non-dimensionalizing the above equation one can clearly see the influencing parameters of the problem which form the basis to characterize such a problem. Normalizing

U, t, x_2 with U_0, $1/\omega$ and y one obtains:

$$\frac{\partial U^*}{\partial t^*} = \frac{\nu/\omega}{y^2}\frac{\partial^2 U^*}{\partial x_2^{*2}} \tag{3}$$

where $U = U^* U_0$, $t = t^*/\omega, x_2 = x^* y$ and y denotes the dimensional distance from the flat plate. It can be noted that the solution of the above partial differential equation is a function of the dimensionless coefficient of the diffusive term (right hand side term) of eq. (3), i.e. $St = \frac{y^2}{\nu/\omega}$, where St is the Stokes number. From the similarity analysis one can readily learn that for a well spatially resolved CFD simulation, the numerical Stokes number should be less than one, i.e.:

$$\begin{aligned} St_{\text{num}} = \frac{\Delta x_2^2}{\nu/\omega} &< 1 \\ \Rightarrow \Delta x_2 &< \sqrt{\frac{\nu}{\omega}} \end{aligned} \tag{4}$$

which means that the thickness of the cell next to the wall should be lesser than the magic ratio $\sqrt{\nu/\omega}$. Futher, the domain length in x_2 direction is given by

$$y >> \sqrt{\frac{\nu}{\omega}}. \tag{5}$$

It can also be noted from the above analysis that the characteristic distance for an oscillatory problem is in fact scaled with $\sqrt{\nu/\omega}$. That is the propagation distance of the oscillatory motion of the wall increases with increasing viscosity and decreasing frequency and vice-versa.

To integrate eq. (2) for all $t > 0$, the following boundary conditions (B.C.) are employed:

$$\begin{aligned} U = U_0 \sin(\omega t) \quad &\text{at} \quad x_2 = 0 \\ U = 0 \quad &\text{at} \quad x_2 = \infty >> \sqrt{\nu/\omega} \end{aligned} \tag{6}$$

For the initial condition, the whole system is considered to be stationary, i.e. for $t \leq 0$ and $U = 0$. Finally, on integrating eq. (2) one gets a solution [1]:

$$U = U_0 e^{-\sqrt{\frac{St}{2}}} \cos(\omega t - \sqrt{St/2}) \tag{7}$$

In the argument of the cosine function, the dimensionless quantity "$-\sqrt{St/2}$" is a measure of phase shift between the fluid motion and the oscillatory plate.

Now the same problem is solved numerically for the ratio ν/ω equal to be 10^{-3}. The oscillation frequency was set equal to one, so that the length scale $\sqrt{\nu/\omega}$ can directly be varied via kinematic viscosity only. The numerical grid, as shown in Fig. 2, is constructed such that the for first few cell near the plate

$$\delta y \approx \sqrt{10^{-3}} \approx 0.02 \text{ m}. \tag{8}$$

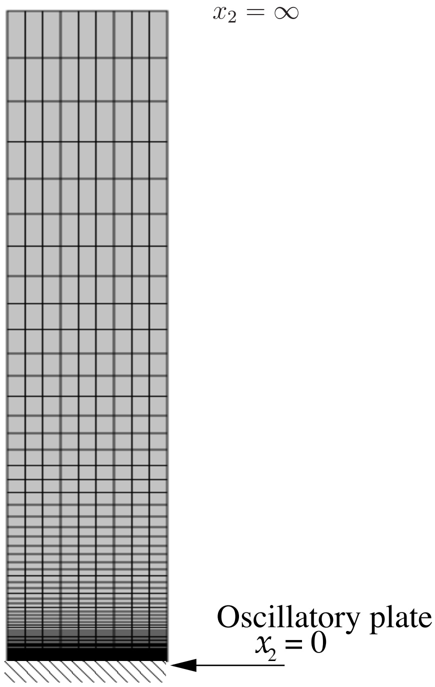

Figure 2: A side view of the numerical grid.

The height of the fluid above the plate was taken to be 4 m >> $\sqrt{\nu/\omega}$. The problem is in fact a one-dimensional problem, i.e. the velocity varies only in x_2 direction. Therefore in all other direction either symmetry or periodic boundary conditions are employed.

For the numerical simulations each cycle contains ten steps and the results obtain at the end of the cycle are plotted in Fig. 3. A reasonably good agreement can be found between the analytical and numerical results.

2.2 Oscillatory Tube

The present test case deals with the circular tube which is oscillating around its axis in the azimuthal direction in order to measure the viscosity of the fluid by the effect of shear force exerted by the fluid on its walls. A schematic view of the problem is shown in Fig. 4 where the tube with radius $R = 8.8$ mm oscillates with a frequency $f = 725$ Hz. The pipe oscillates in the tangential (θ) direction with an amplitude angle of $\Theta_0 = 1°$. For such a configuration, the velocity of the oscillating tube is prescribed as a sine function, i.e.:

$$U_t(r = R, t) = U_0 \sin(\omega t) = R\Theta_0\omega \sin(\omega t) \tag{9}$$

where ω is the angular frequency of the oscillations, r the radial coordinate direction and U_t the tangential velocity in the azimuthal direction θ.

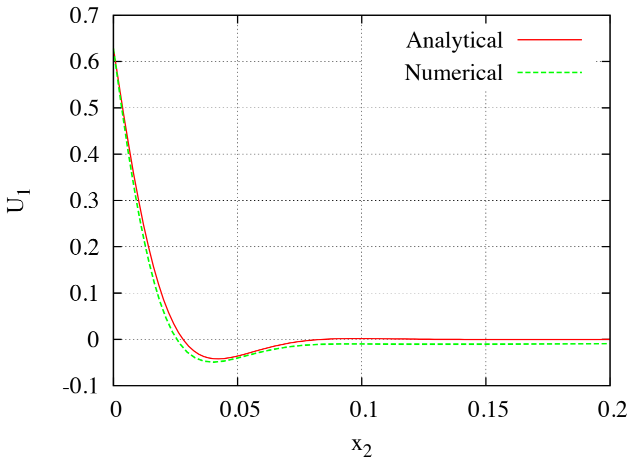

Figure 3: Comparison of periodic numerical and analytical (eq. (7)) solutions for the case of flow induced by oscillatory flat plate underneath a semi-infinite fluid at the end of a cycle.

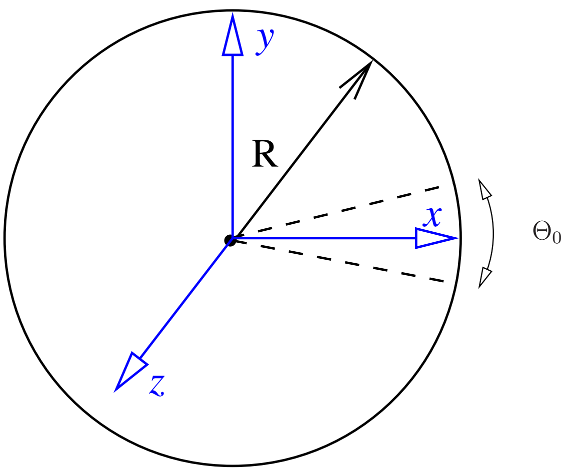

Figure 4: A schematic cross-sectional view of the oscillating around z axis.